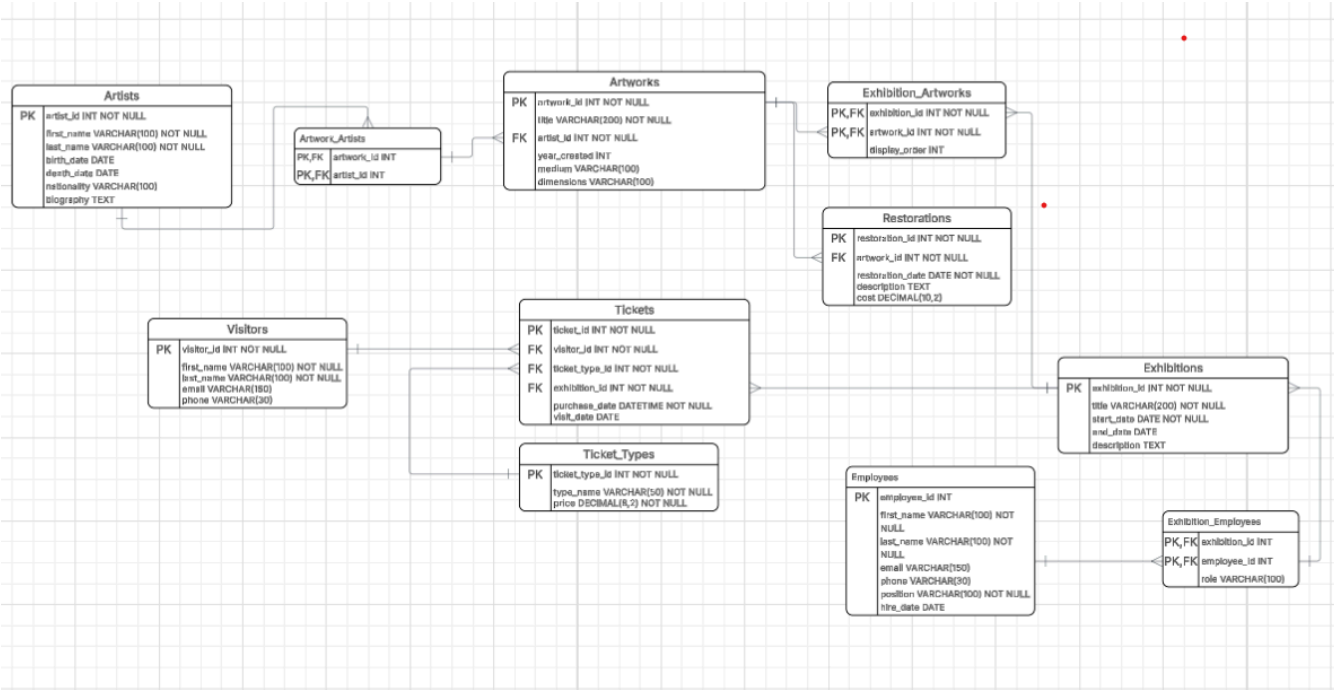
1.1 Logical scheme



1.2. Objects

Artists

Information about artists who created artworks displayed in the museum: first and last names, birth and death dates, nationality, and biography.

Table Name	Field name	Field Description	Data Type / Constraints
Artists	artist_id	Unique artist identifier	INT, NOT NULL, PK
	first_name	Artist first name	VARCHAR(100) NOT NULL
	last_name	Artist last name	VARCHAR(100) NOT NULL
	birth_date	Artist birth date	DATE
	death_date	Artist death date	DATE
	nationality	Artist nationality	VARCHAR(100)
	biography	Artist biography	TEXT

Artist_Artworks

Junction table that represents many-to-many relationship between artworks and exhibitions.

Table Name	Field name	Field Description	Data Type / Constraints
Artist_ Artworks	artist_id	Artist identifier	INT PK FK
	artwork_id	Artwork identifier	INT PK FK

Artists ↔ Artworks: many-to-many relationship. One artist can create many artworks, one artwork can be created by several artists.

Artworks

Information about artworks owned or displayed by the museum: the artwork title, the id of the artist who created the artwork, creation year, medium, and dimensions.

Table Name	Field name	Field Description	Data Type / Constraints
Artworks	artwork_id	Unique artwork identifier	INT NOT NULL PK
	title	Artwork title	VARCHAR(200) NOT NULL
	artist_id	Artist identifier	INT NOT NULL FK
	year_created	Year artwork was created	INT
	medium	Material or technique used	VARCHAR(100)
	dimensions	Artwork dimensions	VARCHAR(100)

Artworks → Restorations: one-to-many relationship. One artwork can have many restorations over time.

Restorations

Information about restoration work performed on artworks: the id of the artwork, restoration date, description, and the cost.

Table Name	Field name	Field Description	Data Type / Constraints
Restorations	restoration_id	Restoration identifier	INT NOT NULL PK
	artwork_id	Artwork identifier	INT NOT NULL FK
	restoration_date	Date of restoration	DATE NOT NULL
	description	Restoration description	TEXT
	cost	Restoration cost	DECIMAL(10,2)

Restorations → Artworks: many-to-one relationship. Many restorations can be made for one artwork.

Exhibition_Artworks

Junction table that represents many-to-many relationship between artworks and exhibitions.

Table Name	Field name	Field Description	Data Type / Constraints
Exhibition_ Artworks	exhibition_id	Exhibition identifier	INT NOT NULL PK FK
	artwork_id	Artwork identifier	INT NOT NULL PK FK
	display_order	Order of artwork display	INT

Artworks ↔ Exhibitions: many-to-many relationship. One painting can appear in many exhibitions, one exhibition contains many artworks.

Exhibitions

Information about exhibitions organized by the museum: the exhibition title, start and end dates, and description.

Table Name	Field name	Field Description	Data Type / Constraints
Exhibitions	exhibition_id	Unique exhibition identifier	INT NOT NULL PK
	title	Exhibition title	VARCHAR(200) NOT NULL
	start_date	Exhibition start date	DATE NOT NULL
	end_date	Exhibition end date	DATE
	description	Exhibition description	TEXT

Exhibitions → Tickets: one-to-many relationship. One exhibition can have many tickets for it.

Exhibition_Employees

Junction table that represents many-to-many relationship between curators and exhibitions.

Table Name	Field name	Field Description	Data Type / Constraints
Exhibition_ Curators	exhibition_id	Exhibition identifier	INT PK FK
	employee_id	Employee identifier	INT PK FK
	role	Employee role in exhibition	VARCHAR(100)

Exhibitions ↔ Employees: many-to-many relationship. One exhibition can have multiple employees, one employees can manage multiple exhibitions.

Employees

Information about employees working in the establishment: first and last names, email address, phone number, position and hire date.

Table Name	Field name	Field Description	Data Type / Constraints
Employees	employee_id	Employee identifier	INT PK
	first_name	Employee first name	VARCHAR(100) NOT NULL
	last_name	Employee last name	VARCHAR(100) NOT NULL
	email	Employee email	VARCHAR(150)
	phone	Employee phone	VARCHAR(30)
	position	Employee work position	VARCHAR(100) NOT NULL
	hire_date	Employee hire date	DATE

Tickets

Information about tickets purchased by visitors to attend exhibitions: the id of the visitor, the id of the ticket type, the id of exhibition, purchase date, and visit date.

Table Name	Field name	Field Description	Data Type / Constraints
Tickets	ticket_id	Unique ticket identifier	INT NOT NULL PK
	visitor_id	Visitor identifier	INT NOT NULL FK
	ticket_type_id	Ticket type identifier	INT NOT NULL FK
	exhibition_id	Exhibition identifier	INT NOT NULL FK
	purchase_date	Ticket purchase date	DATETIME NOT NULL
	visit_date	Planned visit date	DATE

Tickets → Exhibitions: many-to-one relationship. Many tickets can be for one exhibition.

Ticket_Types

Information about different ticket categories and their prices: name of the type, and the price.

Table Name	Field name	Field Description	Data Type / Constraints
Ticket_Types	ticket_type_id	Ticket type identifier	INT NOT NULL PK
	type_name	Type of ticket	VARCHAR(50) NOT NULL
	price	Ticket price	DECIMAL(8,2) NOT NULL

Ticket_Types → Tickets: one-to-many relationship. Many tickets can have one ticket type.

Visitors

Information about people visiting the museum or art gallery: first and last names, email address, and phone number.

Table Name	Field name	Field Description	Data Type / Constraints
Visitors	visitor_id	Visitor identifier	INT NOT NULL PK
	first_name	Visitor first name	VARCHAR(100) NOT NULL
	last_name	Visitor last name	VARCHAR(100) NOT NULL
	email	Visitor email	VARCHAR(150)
	phone	Visitor phone	VARCHAR(30)

Visitors → Tickets: one-to-many relationship. One visitor can buy many tickets, one ticket belongs to one visitor.

Example with data

Artists

artist_id	first_name	last_name	birth_date	death_date	nationality
1	Pablo	Picasso	1881-10-25	1973-04-08	Spanish
2	Vincent	van Gogh	1853-03-30	1890-07-29	Dutch
3	Leonardo	da Vinci	1452-04-15	1519-05-02	Italian
4	Claude	Monet	1840-11-14	1926-12-05	French

Artworks

artwork_id	title	artist_id	year_created	medium	dimensions
1	Guernica	1	1937	Oil on canvas	349cm x 776cm
2	Starry Night	2	1889	Oil on canvas	74cm x 92cm
3	Mona Lisa	3	1503	Oil on wood	77cm x 53cm
4	Water Lilies	4	1916	Oil on canvas	200cm x 180cm

Restorations

restoration_id	artwork_id	restoration_date	description	cost
1	2	2024-11-10	Canvas cleaning	500.00
2	3	2024-12-01	Paint restoration	1200.00
3	1	2025-01-15	Frame repair	300.00

Exhibition_Artworks

exhibition_id	artwork_id	display_order
1	4	1
2	3	1
3	1	2
3	2	1

Exhibitions

exhibition_id	title	start_date	end_date	description
1	Impressionist Masters	2025-05-01	2025-06-30	Exhibition of famous impressionist works
2	Renaissance Art	2025-07-01	2025-08-15	Renaissance masterpieces
3	Modern Art	2025-09-01	2025-10-10	Modern art exhibition

Exhibition_Curators

exhibition_id	curator_id	role
1	1	Lead Curator
2	2	Exhibition Manager
3	1	Curator

Employees

employee_id	first_name	last_name	email	phone
1	Emily	Clark	emily@museum.com	87011112222

2	Robert	Davis	robert@museum.com	87013334444

Tickets

ticket_id	visitor_id	ticket_type_id	exhibition_id	purchase_date	visit_date
1	1	1	1	2025-05-10 10:00	2025-05-15
2	2	2	1	2025-05-12 12:00	2025-05-15
3	3	1	2	2025-07-02 11:30	2025-07-05
4	4	3	3	2025-09-01 9:00	2025-09-03

Ticket_Types

ticket_type_id	type_name	price
1	Standard	15.00
2	Student	10.00
3	VIP	30.00

Visitors

visitor_id	first_name	last_name	email	phone
1	Anna	Brown	anna@email.com	87015551111
2	David	Lee	david@email.com	87016662222
3	Maria	Garcia	maria@email.com	87017773333
4	Alex	Johnson	alex@email.com	87018884444